Can a smaller giveaway make a better product launch?

An agent-based decision experiment for a hypothetical campus bottle seller

300
individual customers

60
campaign periods

36
tested policy cells

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Scope: MODEL_SPEC.md; 36 cells include the exploratory three-sample extension.

The seller chooses the launch. Customers decide to buy.

CHOOSE

Price: Rs 400 / 500 / 600

Free bottles: 3 or 12

Who receives them?

SIMULATE

Customers talk to contacts

Opinions and awareness change

Eligible individuals buy once

EVALUATE

Paid buyers, not free owners

Revenue and all bottle costs

Profit and uncertainty

Discussion-contact policy is an additional scenario: random or similar-opinion replacements.

Boundary: one product; one purchase/customer; no inventory shortage or fixed overhead. MODEL_SPEC.md

Synthetic inputs are specified, not fitted to campus data.

300 people, 1,200 ties

Three known groups; stronger initial preference for within-group contacts.

Budgets and value

Belief Beta(2,2)

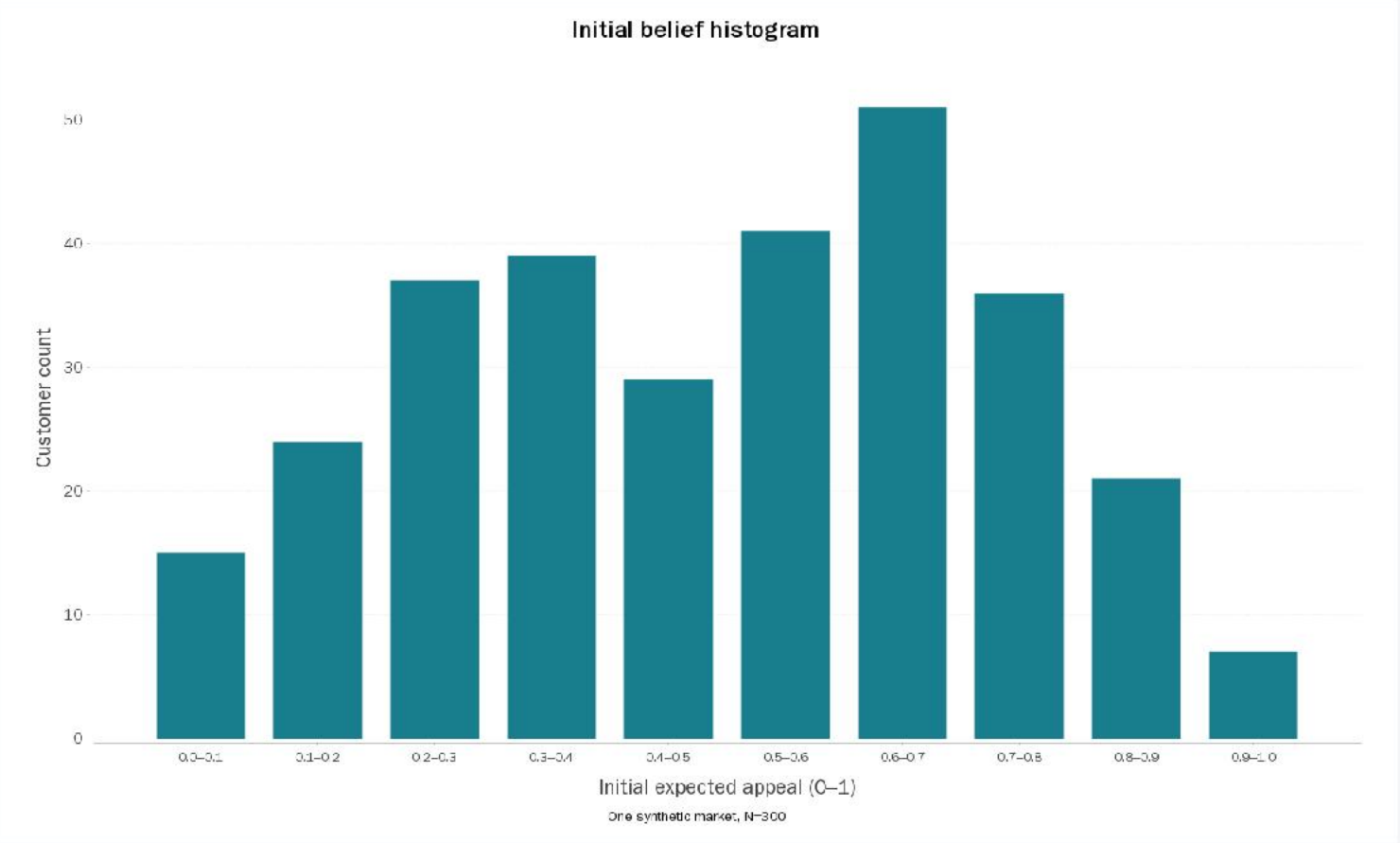
Value scale Rs 500–1,000

Affordability Rs 400–1,000

Example market, rep 1000

Belief mean 0.490; SD 0.235.

Budget mean Rs 710.93.



data/customers_rep1000.csv; sim/analysis/input_statistics.csv. Histogram is one synthetic market.

A conversation changes beliefs. A purchase needs value AND budget.

1	INFORM	An aware contact shares launch news with probability 0.25.
2	DISCUSS	Close beliefs compromise: 0.40 and 0.50 become 0.43 and 0.47.
3	BUY ONCE	Aware + can afford + perceived value ≥ price. Purchases at period end.
4	EXPERIENCE	New owners receive personal feedback. Beliefs and later conversations can change.

Rules: MODEL_SPEC.md. d=0.2; mu=0.3; phi=0.05. Worked opinion example is illustrative.

Established mechanisms; explicit new assumptions; checked code.

LITERATURE

Bounded confidence and adaptive ties

Promotional seeding and diffusion

No claim to invent these mechanisms

OUR ASSUMPTIONS

Price/value/budget threshold

Experience resets and cost accounting

Synthetic graph and attributes

EVIDENCE

16 product checks passed

First AnyLogic default matches Java

Revised code: actual API compile passes

Deffuant et al. 2000; Kozma & Barrat 2008; Delre et al. 2007/2010; Kempe et al. 2003. See LITERATURE_NOTES.md and VALIDATION_REPORT.md.

Compare policies on the same market; confirm on fresh markets.

PILOT

30 separate markets

18 policies

Choose a provisional policy

MAIN + FOLLOW-UP

100 main markets

18 + 18 policies

3,600 evaluations, paired

CONFIRMATION

100 fresh markets

4 fixed comparisons

No search for a new winner

95% intervals: 10,000 bootstrap resamples of markets.

Customers, periods and reused policies are not independent replications.

sim/results/*manifest.json; sim/analysis/confirmation_paired.csv. Exploratory follow-up explicitly distinguished from initial design.

Almost everyone hears. Far fewer buy at the high price.

News is not conversion

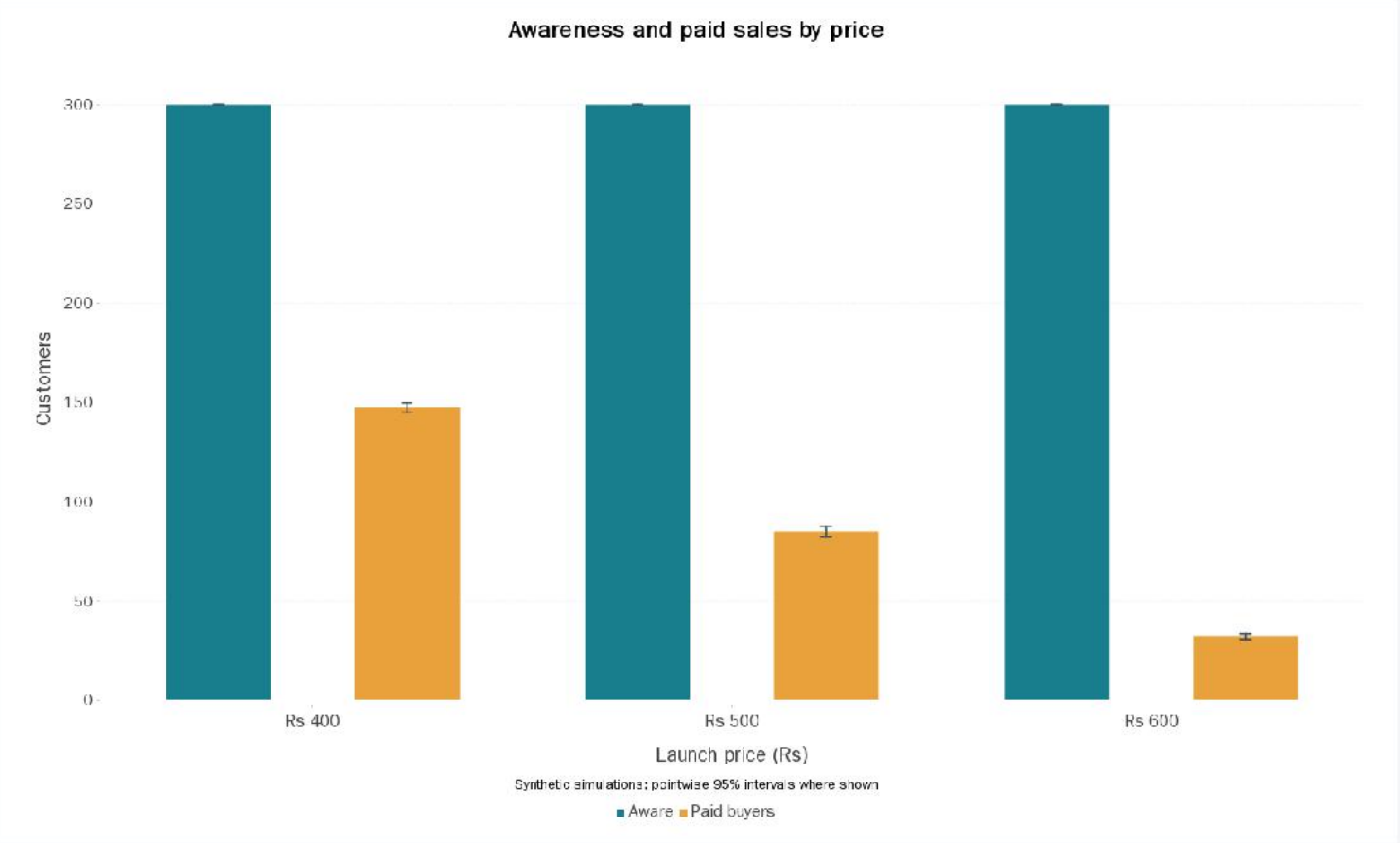
At Rs 600, awareness is nearly 300; mean paid buyers are 32.21.

Margin is not profit

Profit means: Rs 19,100 / 18,213 / 8,274 at the three prices.

Same campaign setup

12 samples; degree targeting; random contact replacement.



sim/analysis/policy_summary.csv; 100 main markets/cell; error bars are pointwise 95% intervals for means.

Nine fewer free bottles can increase profit while reducing total owners.

+ Rs 2,854.50

Fresh-market paired profit gain

95% CI: Rs 2,698.50–3,006

Why it happens

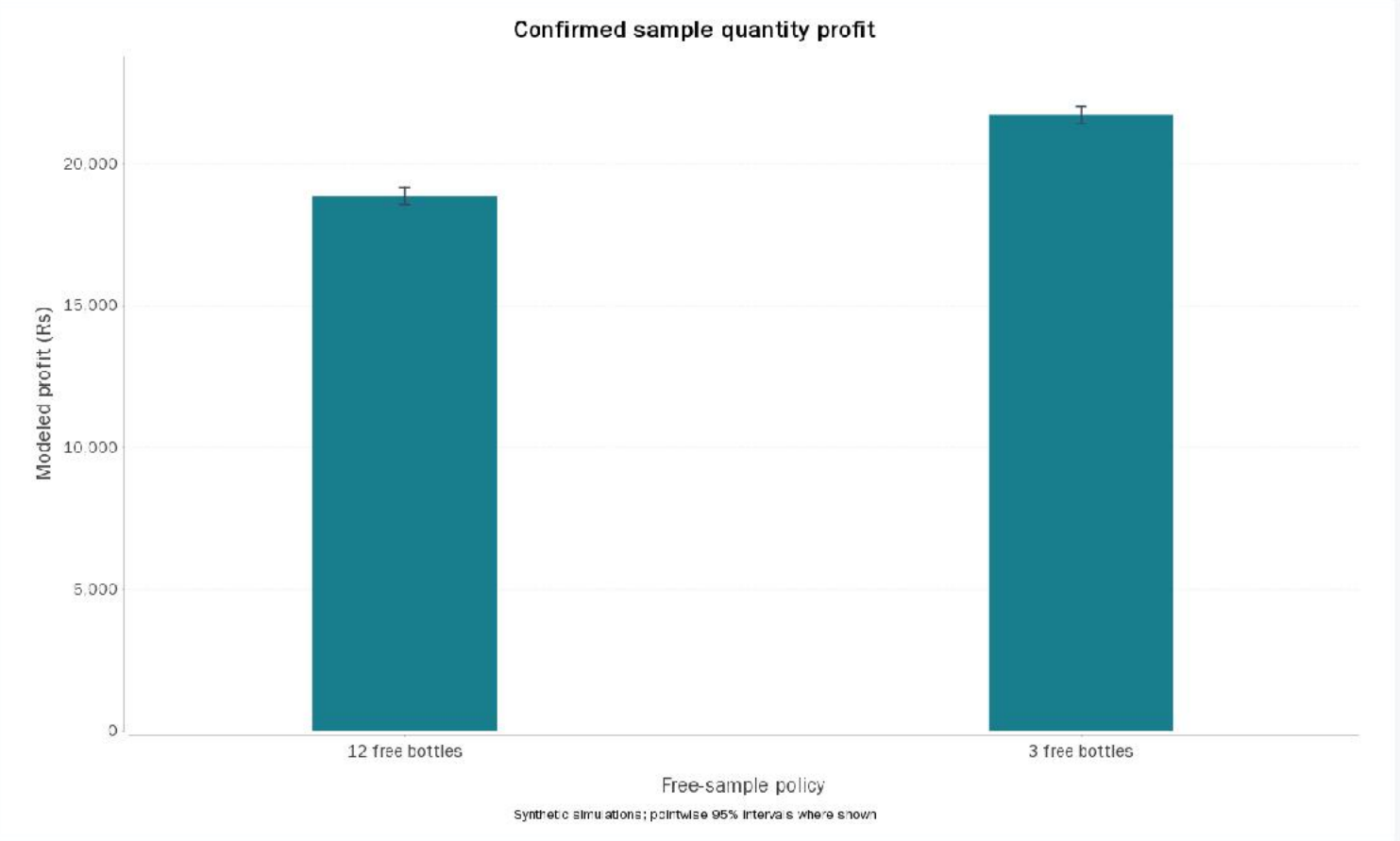
Rs 2,250 less sample cost,

plus margin on about 4 extra sales.

Different objectives differ

Paid sales rise; total owners fall.

Three vs twelve is not a global quantity optimum.



sim/analysis/confirmation_summary.csv and confirmation_paired.csv; 100 fresh paired markets; price Rs 400, degree, h=0.

The sample saving is clear. The influencer premium is uncertain.

Degree over random

+ Rs 31.50

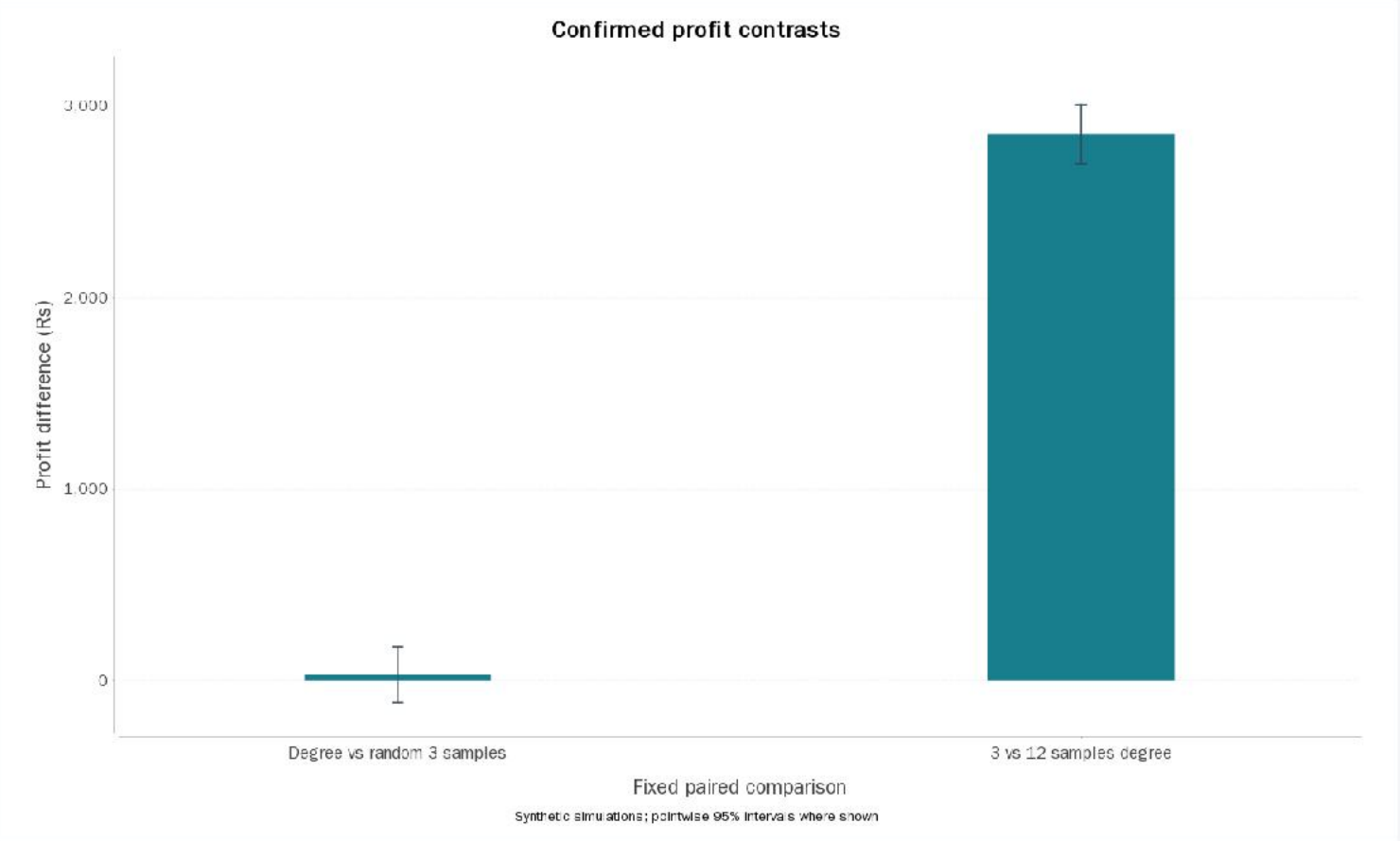
95% CI: −112.50 to +177

Avoid expensive targeting

We have not established a large final-profit benefit from degree information.

Timing can still matter

Strong-group stress: distributed seeding helps early coverage, not clearly final profit.



confirmation_paired.csv; stress_paired.csv. Pointwise intervals; lack of a clear difference is not equivalence.

A surprise exposes a risky assumption: very positive opinions can be ignored.

At narrow confidence

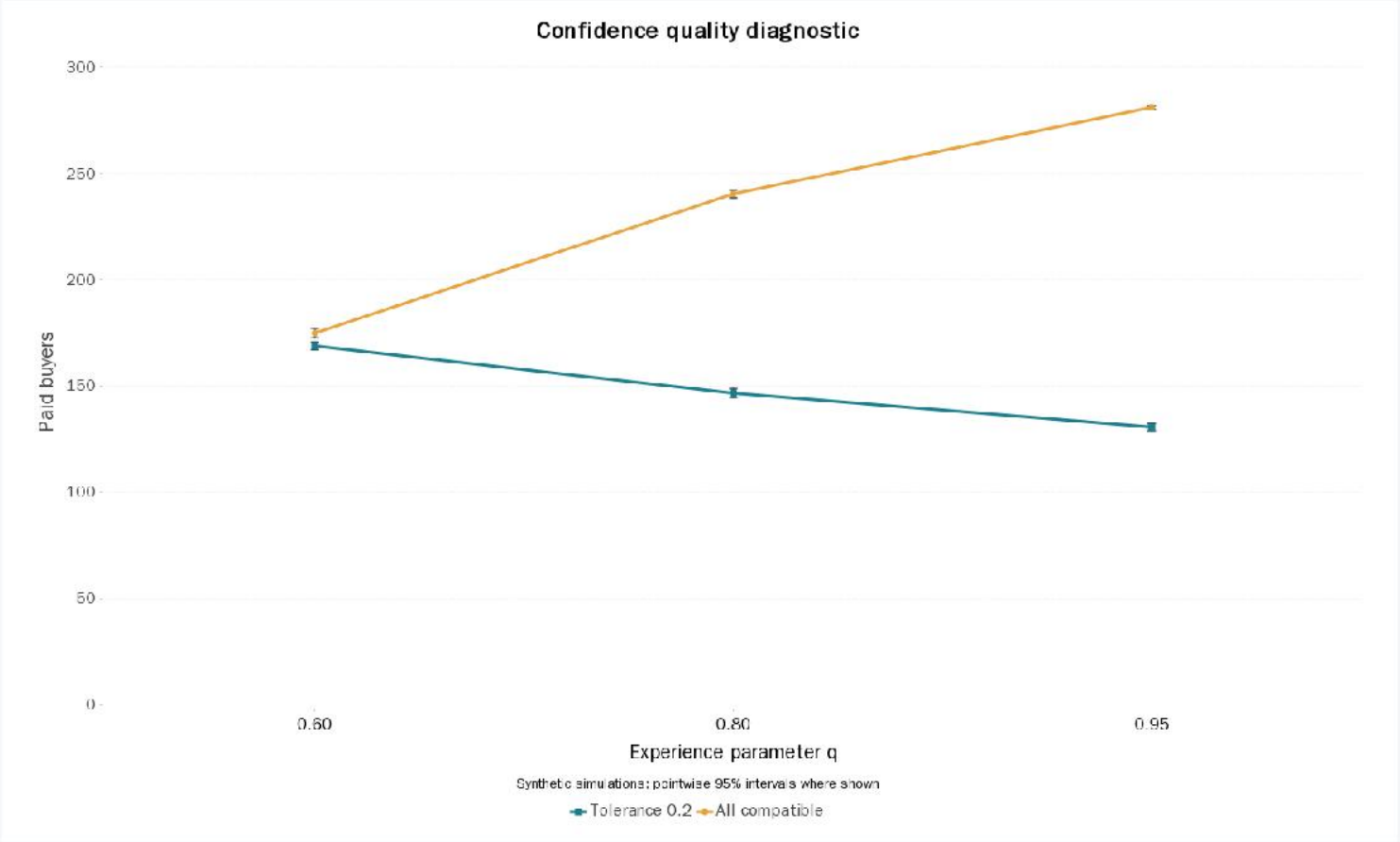
Higher experience mean can reduce simulated sales.

Remove the gate

With all opinions compatible, the quality effect becomes positive.

Interpretation

Validate or soften the attention rule. Do not recommend worse products.



quality_diagnostic_summary.csv; quality_paired.csv. Exploratory 100 paired markets/setting; fixed graph, Rs 400, 12 samples, degree.

Live: follow one conversation, then check the campaign accounts.

WATCH

Agent colors: awareness and ownership

Selected link: a recent meeting

Opinions: values and distribution

EXPLAIN

Close opinions move together

Distant opinions may change contacts

Buying requires budget and value

CHECK

Default: 148 paid + 12 free

Revenue Rs 59,200 − cost Rs 40,000

Profit Rs 19,200

Run the engine-tested model. Defaults: policyIndex=6, rep=1000. Keep the grid model as fallback.

Approved grid default has real screenshot/CSV evidence. LiveNetwork API/wiring pass; its actual engine run is tracked separately.

Start small. Revisit price with cost. Validate the behavior before deployment.

01 Baseline candidate: Rs 400, three samples. Random targeting remains a practical alternative.

02 Budget is a ceiling, not a spending target. Cost changes can change the preferred price.

03 Synthetic assumptions drive outcomes. The quality reversal is a warning to validate behavior.

Best among tested candidates; no claim of a global optimum or actual-campus sales forecast.

confirmation_paired.csv; all_cost_revaluation.csv; PROJECT_REPORT.md. AI-assisted research, code and drafting.

Profit versus price under the original equal-sample design.



all_policy_summary.csv. 12 samples, h=0; 100 markets; pointwise mean intervals.

The preferred price depends on unit cost.



all_cost_revaluation.csv. 3 samples, degree, h=0; cost-only re-accounting holds customer behavior fixed.

What we did not establish.

- No real demand calibration or empirical sales validation.
- No global optimum: only three prices and two sample quantities.
- No costly network-data collection, competition, inventory or fixed overhead.
- Hard confidence, independent attributes and informed-nonbuyer sharing are assumptions.
- Finite campaign outcomes are not equilibrium claims.
- Pointwise intervals do not protect every comparison simultaneously.
- Actual engine evidence must be kept separate from reference and Java runs.

OPEN_ISSUES.md; VALIDATION_REPORT.md; MODEL_SPEC.md.

Sources and the boundary of our contribution.

Deffuant et al. (2000), Mixing beliefs among interacting agents.

Kozma & Barrat (2008), PRE 77, 016102. doi:10.1103/PhysRevE.77.016102

Delre et al. (2007), JBR 60, 826–835. doi:10.1016/j.jbusres.2007.02.002

Delre et al. (2010), JPIM 27, 267–282. doi:10.1111/j.1540-5885.2010.00714.x

Kempe, Kleinberg & Tardos (2003), Maximizing the Spread of Influence.

Replication-informed extension; seeding and homophilic rewiring are established.

Full reading limits and claim checks are preserved in LITERATURE_NOTES.md.

Direct source links and remaining extraction limits are in the literature notes; no research-priority claim.